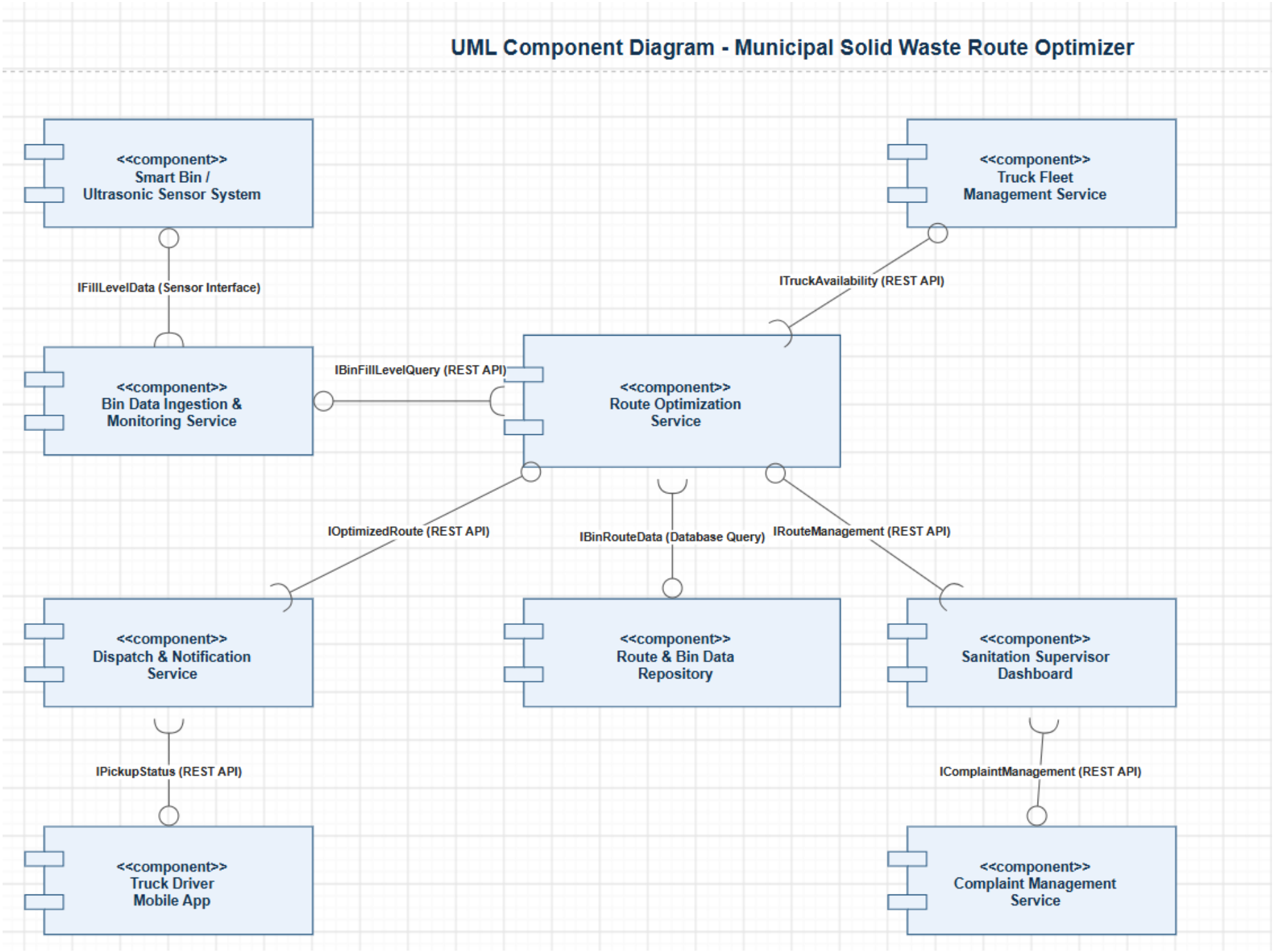

UML Component Diagram - Municipal Solid Waste Route Optimizer
<<component>>
Smart Bin /
Ultrasonic Sensor System
<<component>>
Truck Fleet
Management Service
IFillLevelData (Sensor Interface)
ITruckAvailability (REST API)
<<component>>
Bin Data Ingestion &
Monitoring Service
IBinFillLevelQuery (REST API)
<<component>>
Route Optimization
Service
IOptimizedRoute (REST API)
IBinRouteData (Database Query)
IRouteManagement (REST API)
<<component>>
Dispatch & Notification
Service
<<component>>
Route & Bin Data
Repository
<<component>>
Sanitation Supervisor
Dashboard
IPickupStatus (REST API)
IComplaintManagement (REST API)
<<component>>
Truck Driver
Mobile App
<<component>>
Complaint Management
Service

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Municipal Solid Waste Route Optimizer

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Architecture Selection

We chose Microservices Architecture for the Municipal Solid Waste Route Optimizer. The system is divided into focused services such as Bin Data Ingestion & Monitoring, Truck Fleet Management, Route Optimization, Dispatch & Notification, and Complaint Management. Each service communicates through defined interfaces.

Architectural Style Analysis

Architecture	Suitability for this system
Layered	Clear separation of concerns, but less suitable for independently scaling route optimization and other services.
Microservices	Best fit: independent services, independent scaling, and fault isolation.
Client-Server	Simple centralized control, but may create a scalability bottleneck and single point of failure.

Two Reasons for the Choice

1. **Independent services:** Bin monitoring, fleet management, route optimization, dispatch, and complaint tracking can be developed and maintained separately. This keeps responsibilities clear and makes future changes easier.
2. **Scalability and performance:** The Route Optimization Service can be scaled independently when route-recalculation demand increases. The system must recalculate routes for **50 trucks and 10,000 bins within 45 seconds under peak load**.

Security Advantage

Access to route and operational data can be restricted at the service/interface level using authentication and authorization. This supports the requirement that only authorized users can access operational information and prevents unauthorized changes.

Performance Benefit

Route calculations can run as an independent service without requiring the entire application to scale. Additional resources can therefore be focused on route optimization during peak workloads, helping the system meet its 45-second recalculation target.

Conclusion

Microservices Architecture is appropriate because it provides modularity, independent scalability, clear service responsibilities, and better fault isolation for the waste-management workflow.